FIITJEE

ALL INDIA INTEGRATED TEST SERIES

HALF COURSE TEST – V

JEE (Main)-2021

ANSWERS, HINTS & SOLUTIONS

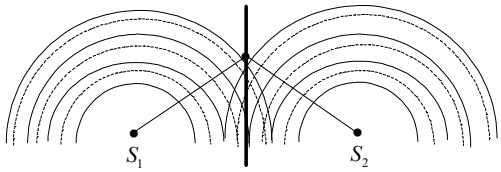
Physics

PART – I

SECTION – A

1. A

Sol.



2. A

3. D

Sol. $T = mg - v\rho g$

4. C

5. A

Sol. Let the frequency of first fork = n
So the frequency of last fork =3n
Each fork gives 4 beats so the frequencies are
$n, n+4, n+8.....3n$
$3n = n + (56-1)d$
$n = 110$

6. C

7. C

Sol. $\frac{dm}{dt}(\Delta v) = \rho a v (2v)$

8. B

Sol. Work done along path 1 is less than that of along 2 and change in internal energy is same.

9. A

Sol. $v_{BA} = v_B - v_A = 20 - 10 = +10 m/s$

$a_{BA} = a_B - a_A = -1 - 4 = -5 m/s^2$

v_{BA} will be zero after $\frac{10}{5}$ or 2s

10. A

Sol. $a_1 = \frac{10}{10} t = t, a_2 = 10 m/s^2$

$a_{CM} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2} (m_1 = m_2)$

$= \frac{(a_1 + a_2)}{2} = \left(5 + \frac{t}{2}\right)$

At $= 0, a_{CM} = 5 m/s^2$ and $t = 10s, a_{CM} = 10 m/s^2$

So, the graph is a straight line with intercept $5 m/s^2$ and slope $\frac{1}{2} m/s^2$

11. A

Sol. Equilibrium of m gives

$T = mg \quad (T = \text{Tension in string})$

Net torque about point of contact of spool should be zero, Hence,

$(2R)(Mg \sin\alpha) = TR$

Or $2Mg \sin\alpha = mg$

Or $m = 2M \sin\alpha$

12. B

Sol. Sx = I α

$\alpha = \left(\frac{F}{I}\right) x$

Therefore acceleration of centre of mass is propartional to x

13. C

Sol. Angular momentum will remain conserve about point O

$L_1 = L_2$

$2mV_0 r_0 = 2mV\left(r_0 \cos^2\theta\right)$

14. A

Sol. $KE_i + U_i = KE_f + U_f$

$0 + m_1 g\left(R - R\cos\theta\right) = \frac{1}{2} m_1 v^2 + 0$

$$\sum F_n = T - m_1 g = m_1 \frac{v^2}{R}$$

$$T = m_2 g$$

From equation (1), (2) and (3), we get

$$m_2 g = m_1 g + m_1 \frac{2gR(1-\cos\theta)}{R}$$

$$\cos\theta = \frac{3m_1 - m_2}{2m_1}$$

$$= \frac{3\times 3 - 6}{2\times 3} = \frac{1}{2} = \theta = 60°$$

15. A

Sol. $V_2 = \frac{-GM}{R} at\ the\ centre$

$$\frac{dv}{dt} = -\left(\frac{-GM}{R^2}\right)\frac{dR}{dt}$$

$$\frac{dv}{dt} = \frac{GM}{R^2}\frac{dR}{dT}$$

Increases

16. C

Sol. Spring should be cut at the centre of mass of system

$$X_{com} = \frac{mxo + m\ell}{m+M}$$

$$X = \frac{M\ell}{m+M} from\ m$$

17. A

Sol. Angular frequency of SHM is equal to angular velocity of particle $\omega = V/R$.

18. D

Sol. $$ms\frac{\Delta\theta}{dt} = \frac{KA(\theta_1 - \theta_2)}{\ell}$$

$$\frac{\Delta t}{\Delta\theta} \equiv \frac{21}{300} \Rightarrow t=70ms$$

19. C

Sol. $P=kV$

$PV^{-1}=k$

$PV^n = constant \Rightarrow n\text{-}1$

$$C=C_v + \frac{R}{1\text{-}n} = C_v + \frac{R}{2}$$

$$= \frac{5R}{2} + \frac{R}{2} = 3R$$

20. A

Sol. $W = \int PdV$ and $P \propto V$

SECTION – B

21. 5

22. 3

SECTION – C

23. 00000.35

Sol. $T \cos 30^0 = f = \mu N$

$T \sin 30^0 = 8$

$\tan 30^0 = \dfrac{8}{\mu N}$

$\mu = \dfrac{8}{N \tan 30^0} = \dfrac{8\sqrt{3}}{40} = 0.35$

24. 00004.00

Sol. $EP = \text{semi major axis} = \dfrac{3+5}{2}$

$EP = 4$

25. 00010.00

Sol. The cylinder begins to slide when vertical line through C.G. passes beyond the base of cylinder.

$\tan\theta = \dfrac{r}{\left(\dfrac{h}{2}\right)} = 1$

Or $r = \dfrac{h}{2} = 10\text{cm}$

Chemistry

PART – II

SECTION – A

26. C
Sol. Zeroes between two non-zero digits are significant

27. B
Sol. H attached with sp-hybridised carbon is more acidic

28. D
Sol. Down the group ionic radius increases

29. D
Sol. XeF_6 and XeO_3 is has non-symmetrical structure, so non-zero dipole moment

30. D
Sol. 3° carbocation is more stable than bridge-head carbocation

31. B
Sol. Down the group, inert pair effect increases due to the presence of f- subshell electrons

32. C
Sol. In option C each carbon has different groups

33. A

34. C
Sol. π-bond are $\pi 2p_x$ and $\pi 2p_y$, if z is internuclear axis

35. A

36. A
Sol. $N_1V_1 = N_2V_2$

37. C
Sol. $$\frac{\left[H^+\right]}{\left[OH^-\right]} = \sqrt{\frac{K_a.C}{K_b.C'}} \qquad \left[\because K_a = K_b\right]$$
$$\Rightarrow \frac{\left[H^+\right]}{\left[OH^-\right]} = \sqrt{\frac{C}{C'}}$$

38. D
Sol. In graphite, each c-atom is sp^2 hybridised and exists in a sheet of hexagons (each c-atom is bonded to 3 others). This means that each carbon atom forms 3σ bond with its neighbours and then shares a π bond. So overall $\frac{4\,bonds}{3\,atoms} = \frac{4}{3} = 1.33$

39. C

Sol. $\ln\frac{K_2}{K_1}=\frac{\Delta H^\circ}{2.303R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$

40. D

Sol. $C_V=3R=25$

41. C

Sol. Above critical temperature, it will show ideal gas behaviour so $Z=1$

42. B

Sol. Thiophene has more tendency of EAS

43. D

Sol. Fact

44. D

Sol. One of the carbon has same substituents attached to it.

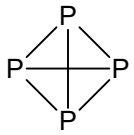
45. B

Sol. Presence of f-centre does not make the lattice electrically neutral due to the presence of electrons.

SECTION – B

46. 6

Sol.



47. 4

Sol. Specific rotation of (S)-enantiomer = - 48

$$\Rightarrow 25\%\,of\,(S)+75\%\,of\,(R)=\frac{25}{100}(-48)+\frac{75}{100}(48)=6x=24\Rightarrow x=4$$

SECTION – C

48. 00039.30

Sol. $\mathrm{IE}=\frac{-13.6\left(Z_{eff}\right)^2}{n^2}eV$

49. 00542.52

Sol. $\Delta H=\Delta U+\Delta n_g RT\Rightarrow \Delta H=550000-3\times 8.314\times 300=550000-7482.6=542.52\,KJ$

50. 00001.04

Sol. $\mathrm{C_3H_8(g)+O_2(g)\longrightarrow 3CO_2(s)+4H_2O(g)}$

$\mathrm{(y\text{-}x)\quad (5x)\quad (3x)\quad (4x)}$

Mathematics

PART – III

SECTION – A

51. D

Sol. Differentiate both sides w.r.t. x

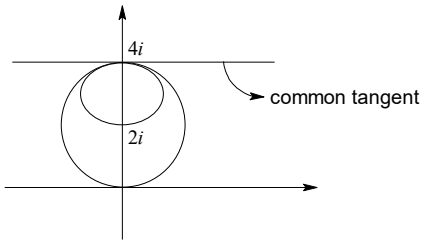
$$n(1+x)^{n-1} = C_1 + 2C_2 + 3C_3x^2 + ...+ nC_nx^{n-1}$$

Substrate $x = 1$

$$n.2^{n-1} = C_1 + 2C_2 + 3C_3 + ...+ nC_n$$

52. C

Sol.



53. B

Sol. The given expression is the coefficient of x^4 in

$${}^4C_0(1+x)^{404} - {}^4C_1(1+x)^{303} + {}^4C_2(1+x)^{202} - {}^4C_3(1+x)^{101} + {}^4C_4$$

$$= \left((1+x)^{101} - 1\right)^4$$

$$= \left({}^{101}C_1x + {}^{101}C_2x^2 + ...\right)^4$$

$$= (101)^4$$

54. D

Sol. T_n of the series is $\frac{2n^2+n+1}{n!} = \frac{2n}{(n-1)!} + \frac{1}{(n-1)!} + \frac{1}{n!} = \frac{2}{(n-2)!} + \frac{3}{(n-1)!} + \frac{1}{n!}$

$$= 2e + 3e + (e-1)$$

$$= 6e - 1$$

55. D

Sol. $-\omega, -\omega^2$ satisfy $x^n + x - 1 = 0$

$\Rightarrow n = 6k - 1$

or $n = 6\lambda + 5$

56. B

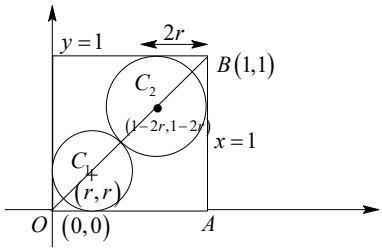
Sol. Here $x = \frac{y^n}{8-6y}$, but x is non – negative

⇒non – negative value of y can be 0 and 1.

Now $y = 0 \Rightarrow x = 0$ and $y = 1 \Rightarrow x = \frac{1}{2}$

57. C

Sol. C_1, C_2 *lie on* $y = x$ *line*



$$OB = OC_1 + C_1C_2 + C_2B$$

$$\sqrt{2} = r\sqrt{2} + 3r + 2r\sqrt{2}$$

$$r = \frac{\sqrt{2}}{3\left(\sqrt{2}+1\right)}$$

58. C

Sol. The equation of the bisectors of the angle between the lines $x^2 - 2axy - y^2$ is

$$\frac{x^2 - y^2}{1-(-1)} = \frac{xy}{-a}$$

$$\Rightarrow ax^2 + 2xy - ay^2 = 0$$

This is the same as $x^2 - 2bxy - y^2 = 0$

$$\therefore \frac{a}{1} = \frac{1}{(-b)} = \frac{-a}{-1}$$

$$\Rightarrow -ab = 1$$

$$\Rightarrow ab + 1 = 0$$

59. C

Sol. Given chord is focal chord, therefore angle between normals is $\frac{\pi}{2}$.

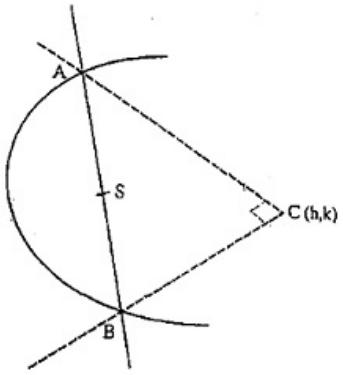
Let C be (h, k)

Equation of normal is

$y = mx - 2am - am^3$ .....(i)

$\Rightarrow$ put (h, k)

$\therefore\ m^3 + (2 - h)m + k = 0$ .....(ii)



Let m_1, m_2, m_3 are roots

$\therefore\ m_1 m_2 m_3 = -k$

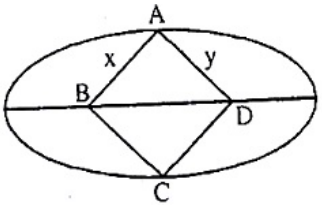
Put in (ii) $k^3 + k(2-h) + k = 0$

$\Rightarrow k = 0$ or $k^2 + 3 - h = 0$.

60. A

Sol. $(x+y)^2 = x^2 + y^2 + 2xy = 25$

$\Rightarrow\ x + y = 5\ \Rightarrow\ 2a = 5$



Now $2ae = \sqrt{13}\ \Rightarrow\ e = \frac{\sqrt{13}}{5}$

$\therefore\ b^2 = \frac{25}{4}\left(1 - \frac{13}{25}\right) = 3\ \Rightarrow\ b = \sqrt{3}$

Area $= \pi ab = \pi \times \frac{5}{2} \times \sqrt{3}$.

61. D

Sol. $-3 < \frac{k^2}{4} - 4 < 0$

62. B

Sol. $t_2 = -t_1 - \frac{2}{t_1}, t_3 = -t_2 - \frac{2}{t_2}$

$\therefore 3t_1 + 13t_2 + 9t_3 = 0$

$\Rightarrow t_1 = 2$

63. D

Sol. The number of non-negative integer solution of $2x + y + z = 21$

64. A

Sol. For points of intersection,

$x^4 + 1 = 2x^2 - \sin^2 \pi x,\ (x^2 - 1)^2 = -\sin^2 \pi x \quad \Rightarrow \quad x = \pm 1$

$$\Rightarrow \int_{-1}^{1} (x^4 + 1 - 2x^2 + \sin^2 \pi x)dx = 2\left|_0^1 \left(\frac{x^5}{5} + x - \frac{2x^3}{3} + \frac{1}{2}x - \frac{\sin 2\pi x}{4\pi}\right)\right.$$

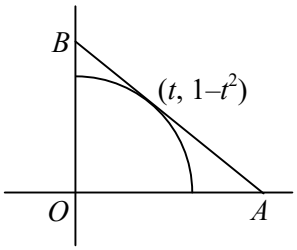
$$= \frac{31}{15}$$

65. B

Sol. Equation of tangent = $\frac{2tx}{1+t^2} + \frac{y}{1+t^2} = 1$

So, area of $\Delta = \frac{1}{2} \times \left(\frac{1+t^2}{2t}\right) \times (1+t^2)$

$= \frac{1}{4}\frac{(1+t^2)^2}{t}$



To minimize $\frac{d}{dt}\frac{(1+t^2)^2}{t} = 0$

$\Rightarrow \quad \frac{(1+t^2)}{t^2}(3t^2-1) = 0$

So, minimum exist at $x = \frac{1}{\sqrt{3}}$

Area = $\frac{4\sqrt{3}}{9}$

66. B

Sol. $a_n = \sqrt{2+\sqrt{2+\sqrt{2}+....}}$ having n radical signs. Then $a_1 = \sqrt{2} = 2\cos\frac{\pi}{4} or\ 2\sin\frac{\pi}{4}$

$\therefore\ a_1$ may be true for option (A) as well as (B)

Again, $a_2 = \sqrt{2+\sqrt{2}} = 2.\frac{\sqrt{2+\sqrt{2}}}{2} = 2\cos\frac{\pi}{8}$

$\therefore\ a_2$ is true for option (B)

$\therefore$ we will check the result for option (B)

Let $\underbrace{\sqrt{2+\sqrt{2+\sqrt{2}+...}}}_{m\ roots} = 2\cos\left(\frac{\pi}{2^m+1}\right)$, for some $m \geq 1$

Now $\underbrace{\sqrt{2+\sqrt{2+\sqrt{2}+...}}}_{(m+1)roots} = \frac{2+\sqrt{2+\sqrt{2+\sqrt{2}+....}}}{m\ roots}$

$$= \sqrt{2+2\cos\left(\frac{\pi}{2^{m+1}}\right)}$$

$$\sqrt{2.2\cos^2\left(\frac{\pi}{2^{m+2}}\right)}$$

$$= 2\cos\left(\frac{\pi}{2^{m+2}}\right)$$

$$= 2\cos\left(\frac{\pi}{2^{(m+1)+1}}\right)$$

$\therefore$ the result is true for n = m+1. Hence, by induction, option (B) is the correct option.

67. C

Sol. The equation of the bisectors of the angle between the lines $x^2 - 2axy - y^2$ is

$\frac{x^2-y^2}{1-(-1)} = \frac{xy}{-a}$

$\Rightarrow ax^2 + 2xy - ay^2 = 0$

This is the same as $x^2 - 2bxy - y^2 = 0$

$\therefore \frac{a}{1} = \frac{1}{(-b)} = \frac{-a}{-1}$

$\Rightarrow -ab = 1$

$\Rightarrow ab + 1 = 0$

68. C

Sol. Since, a, b, c denote the sides of a triangle

$\therefore a > 0, b > 0, c > 0$

We have $a = z^2 + z + 1$

$= \left(z + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 > 0 \forall z$

$b > 0 \Rightarrow 2z + 1 > 0$

$\Rightarrow z > -\frac{1}{2}$

$c > 0 \Rightarrow z^2 - 1 > 0 \Rightarrow z > 1 or\, z < -1$

Hence, a, b, c are positive for $z > 1$

Now $a - b = \left(z^2 + z + 1\right) - \left(2z + 1\right)$

$= z^2 - z$

$= z\left(z - 1\right) > 0 \; for \; z > 1$

$a - c\left(z^2 + z + 1\right) - \left(z^2 - 1\right)$

$= z + 2 > 0$ for $z > 1$

$\therefore$ a is the largest side of the given triangle.

$\Rightarrow$ the angle opposite to it is the largest angle.

Now, $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$= \frac{\left(2z + 1\right)^2 + \left(z^2 - 1\right)^2 - \left(z^2 + z + 1\right)^2}{2\left(2z + 1\right)\left(z^2 - 1\right)}$

$= \frac{\left(1 - z^2\right)}{2\left(z^2 - 1\right)}$

$= -\frac{1}{2}$

$A = 120°$

69. B

Sol: $\alpha + \beta = \frac{m^2 - 2m}{m^2}, \alpha\beta = \frac{3}{m^2}, \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{3}$

$\Rightarrow (\alpha+\beta)^2 = \frac{10}{3}\alpha\beta \Rightarrow \left(1-\frac{2}{m}\right)^2 = \frac{10}{m^2}$

$\Rightarrow m^2 - 4m - 6 = 0 \Rightarrow m_1 + m_2 = 4, m_1 m_2 = -6$

70. D

Sol. $k = \log_2 \log_3 5^{\lambda} \Rightarrow 2^{-k} = \log_5 3^{\frac{1}{\lambda}}$

$\Rightarrow 1 < 25^{\log_5 3^{1/\lambda}} < 3$

$\Rightarrow \lambda > 2$

SECTION – B

71. 3

72. 1

Sol. Base of logarithm > 0, ≠ 1

$\therefore$ $x^2 + 6x + 8 > 0$ and $2x^2 + 2x + 3 > 0$

$\therefore$ $(x + 4)(x + 2) > 0$ and $\left(x+\frac{1}{2}\right)^2 + \frac{5}{4} > 0$

$\Rightarrow x \in (-\infty, -4) \cup (-2, \infty)$ ...(i)

and $x^2 - 2x > 0$

then $x(x - 2) > 0$

$\therefore$ $x \in (-\infty, 0) \cup (2, \infty)$ ...(ii)

Then the given equation reduced in the form

$\log_{2x^2+2x+3} (x^2 - 2x) = (x^2 + 6x + 8)^0$

$\log_{2x^2+2x+3} (x^2 - 2x) = 1$

$\Rightarrow x^2 - 2x = (2x^2 + 2x + 3)^1$

$\Rightarrow x^2 + 4x + 3 = 0$

$\therefore x = -1, -3$

$\because x = -3$ is not satisfied (i) and satisfied (ii)

$\therefore x = -3$ is not a solution of the given equation.

But $x = -1$ satisfies both conditions (i) and (ii)

$\therefore x = -1$, is only solution of the given equation.

SECTION – C

73. 00132.00

Sol. The number of non-negative integer solution of $2x + y + z = 21$

74. 00011.00

Sol. $E = \frac{1}{y}\left[\left(\frac{1+y}{2}\right)^n - \left(\frac{1-y}{2}\right)^n\right]$ where $y = \sqrt{4x+1}$

$= \frac{1}{y}, \frac{1}{2^n}\left[2\left({}^nC_1 y + {}^nC_3 y^3 + ... + {}^nC_n y^n\right)\right]$

For x^5, E should have the term of y^{11}.

75. 00117.00